

PAVAN RAJ R

AI/ML Engineer | Computer Vision Specialist | NLP & Automation

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with industry experience at Robert Bosch GmbH, specializing in computer vision, industrial automation, and NLP systems. Skilled in building end-to-end ML pipelines using Python, OpenCV, TensorFlow, and PyTorch. Passionate about deploying scalable, production-ready AI solutions that reduce manual errors and improve operational efficiency.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, TensorFlow, PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn

Computer Vision: OpenCV, DeepFace, LBPH, Masked Template Matching, NMS, Pypylon SDK

NLP & Speech: NLP Pipelines, SpeechRecognition, Google Gemini API, Context-Aware Generation

Data & BI: Power BI, DAX, SQL, MySQL

Tools & QA: Git, GitHub, Jupyter, PyCharm, VS Code, Selenium WebDriver, REST APIs, Tkinter, PyQt5

WORK EXPERIENCE

AI/ML & Computer Vision Intern | Robert Bosch GmbH

Bangalore | January 2026 – May 2026

- Engineered Glow Plug Inspection System using Masked Template Matching, automating QC on industrial lines and reducing manual inspection errors by ~85%
- Built real-time image processing pipelines (OpenCV + Pypylon SDK); implemented NMS reducing false positives by 30% in production environments
- Designed Tkinter GUI for live inspection, ROI selection, and template management; deployed on factory floor for operator use
- Delivered timestamped CSV logging with PASS/FAIL traceability and multi-template 0°/180° rotation handling for consistent output

EDUCATION

Master of Computer Applications (MCA) — AI, ML & Data Science | S-VYASA Deemed University, Bangalore | 2024 – 2026 (Pursuing)

Bachelor of Science (BSc) — Physics & Computer Science | Surana Autonomous College, Bangalore | 2022 – 2024

PROJECTS

Glow Plug Inspection System — Industrial Automation | Python, OpenCV, NumPy, Tkinter, Pypylon SDK

End-to-end automated vision system: image capture → masked template matching → tray validation → PASS/FAIL logging. ROI filtering + multi-orientation support improved detection precision by 40% over standard methods.

AI Voice Assistant — NLP & Gemini API | Python, NLP, SpeechRecognition, PyQt5, Google Gemini API, Multithreading

Real-time voice assistant with continuous listening, Google Gemini API for context-aware NLP responses, and system automation features.

Multithreaded PyQt5 UI with modular architecture separating GUI, audio, and AI logic for scalability.

Real-Time Face Recognition & Emotion Classification | Python, OpenCV, DeepFace, LBPH, NumPy

Live webcam system combining LBPH face recognition + DeepFace multi-class emotion classification (happy, sad, angry, neutral). Dynamic user registration without full dataset retraining; modular train/inference scripts for clean deployment.

Power BI Sales Dashboard | Power BI, DAX

Built interactive multi-page Power BI dashboard tracking revenue, churn, and KPIs across regions and product categories. DAX measures for YoY growth, rolling averages, and drill-through reports — cut monthly reporting time by 60%.

CERTIFICATIONS

GenAI Powered Data Analytics — Tata Consultancy Services via Forage (May 2026) | EDA, AI-driven delinquency prediction, data storytelling

Operations Industrial Engineer Job Simulation — Siemens via Forage (May 2026) | Time studies, layout optimization

CORE COMPETENCIES

Problem Solving | Critical Thinking | Teamwork & Collaboration | Adaptability | Attention to Detail